Soubhik Sadhu

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Summary

Final-year Computer Science Engineering student with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, DBMS, and Computer Networks. Backend-focused developer experienced in building scalable systems and RESTful APIs using FastAPI and MySQL with performance optimization. Skilled in writing clean, maintainable code and developing reliable, production-ready software solutions.

Education

SRM Institute of Science and Technology — Chennai, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering (AI & ML) — *August 2023 – May 2027*

- CGPA: 9.19/10.0 | Coursework: Data Structures and Algorithms, Machine Learning, Computer Vision, DBMS

Experience

AI/ML Intern — December 2025 – January 2026

Eastern Coalfields Limited (ECL), A Subsidiary of Coal India — *Asansol, West Bengal*

- Built **ECL-SalarySight**, a full-stack payroll ETL and ingestion system (FastAPI, React, MySQL) replacing a manual Excel workflow, handling monthly salary sheets with inconsistent structure across 5,000+ records/month
- Designed the ETL logic for Excel-to-MySQL ingestion, including upsert operations keyed on a composite primary key (employee number + date/month + area) to guarantee duplicate-free records
- Optimized REST APIs to process 10,000+ records in under 5 seconds, with response latency under 200ms

Projects

SkillSync | *FastAPI, React, PostgreSQL, BERT, FAISS, LLM* | **GitHub** — February 2025 – Present

- Built a career intelligence platform (FastAPI, React, PostgreSQL): job matching, resume-driven career path recommendation, coding-platform tracker, resume builder, and LinkedIn optimizer
- Benchmarked BERT-base vs MiniLM (MRR 0.858 vs 0.797); deployed BERT + FAISS for sub-second job matching with a blended score (60% cosine, 25% profile, 15% DSA)
- Developing **InterroX**, an LLM-based mock interview simulator embedded in SkillSync that generates questions from the uploaded resume

OcuSight | *Python, PyTorch, FastAPI, React, TypeScript, Gemini API* | **GitHub** | **IJIRT Published** — March 2026

- Benchmarked ResNet-50, EfficientNet-B4, and DenseNet-121 on the RFMiD dataset (3,200+ fundus images, 46 retinal conditions), achieving 97.8% per-label accuracy
- Trained with BF16 mixed precision and reached 15ms inference latency on an RTX 4060
- Built a confidence-gated ($\geq$70%) pipeline using the Gemini API to generate clinical insight summaries, deployed end-to-end with FastAPI and React/TypeScript
- Published in IJIRT: *"OcuSight: Benchmark Performance vs. Clinical Deployment Reliability in Retinal Multi-Disease Classification"*

AgroWare | *Python, PyTorch, FastAPI, React, PostgreSQL, RAG, TF-IDF* | **GitHub** — January 2026

- Built a multilingual (English, Hindi, Punjabi, Hinglish) farmer-centric **RAG chatbot**: retrieves relevant passages from curated government agricultural datasets (TF-IDF retrieval) and grounds the generated answers in them to reduce hallucination
- Built a plant disease detection pipeline (25 classes, 5 crops, 15k images), benchmarking four CNN backbones and deploying EfficientNet-B0 at 82.4% accuracy
- Built a crop recommendation model (87.4% accuracy) with automated weather/soil data ingestion and parameter-level explainability for its outputs

Technical Skills

Programming Languages: Python, C, C++, JavaScript, TypeScript, SQL
Generative AI & LLMs: RAG, Embeddings, Vector Search (FAISS), LLM, Prompt Engineering
Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, NLP, BERT, Deep Learning, Computer Vision
Web, Backend & Tools: FastAPI, React, REST APIs, Git, Docker, PostgreSQL, MySQL, OpenCV, Jupyter

Achievements

- Completed Amazon ML Summer School (MLSS) 2026
- Published research paper in IJIRT (Volume 12, Issue 12, May 2026)
- Creative Head (Events & Logistics), SIGKDD ACM Student Chapter: organized technical symposiums and hackathons (including HisenHack 2025) for 2,400+ participants